MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) April 7, 2012
Instructions: Attempt <u>any three out of the following four questions</u> on both pages. <u>Mention which three you would like to have graded</u>. Open book and open notes.

1) A researcher wants to compare two methods (M1 vs M2) to teach math. The researcher picks a random sample of 8 students, and assigns at random 4 students to learn under M1, and the other 4 students under M2. At the end of the term, the researcher picks at random 3 algebra problems, and 3 geometry problems. All 8 students are given the same 6 problems. The response variable Y is the number of minutes to solve the problem.

In what follows, let M denote Method, S denote Subject, P denote Problem, and T denote Type of math(algebra vs geometry).

a) Write out the ANOVA table just showing Sources of Variability and degrees of Freedom. and indicate which SV are fixed effects and which are random.

b) To test for Method effect, what MS terms go in numerator and what MS terms go in denominator.

2) Scientists are studying the sizes of radio-electric disturbances known as "ka-chings" that occur on Mars and Pluto. They have obtained 15 measures of ka-chings from Pluto and 10 measures from Mars. Since Mars is closer to Earth, ka-chings of any size on Mars can be detected. However, because Pluto is much further away from Earth, only ka-chings that are larger than 3 units can be measured.

Table of the 15 ordered Ka-ching measures from Mars and 10 ordered measures from Pluto

Mars	**0.1**	**0.3**	**0.5**	**0.7**	**1**	**20**	**200**	**600**	**800**	**10,000**	**100,000**	**250,000**	**500,000**	**2,000,000**	**5,000,000**
Pluto	**5**	**50**	**55**	**1,200**	**2,000**	**5,000**	**25,000**	**40,000**	**1,000,000**	**10,000,000**					

A. It is believed that the median size of a Ka-ching on Mars is 6 units. Statistical test if the data dispute this belief with a Type-1 error of 0.05.

B. Perform the best test <u>within the limits of the available data</u> of whether the central tendency of the size of a Ka-ching is the same for Pluto as for Mars. <u>Remember, that because Pluto is much further away from Earth, only Ka-chings that are larger than 3 units can be measured on Pluto.</u> Again use a type-1 error of 0.05.

3) A student adviser wants to estimate the average number of hours per day Rutgers students worked on outside jobs. The adviser took a random sample of 50 Rutgers students and observed the following amounts.

Hours worked	**frequency**
0	37
1	8
2	2
3	1
4	0
5	0
6	0
7	0
8	2

a) Compute the sample mean and sample standard deviation of the job hours worked per student.
b) Give the standard large sample 95% confidence interval for the average(mean) job hours per Rutgers students. (choose either $z_{.025}=1.96$ or $t_{49,.025} = 2.01$)
c) Is the sample of size 50 large enough to justify your C.I. in part (b)?

4) A snow plow is used each time there is a snow storm. After each use, the snow plow must have its carburetor serviced. <u>If the carburetor is not broken, it costs $50 to service it</u>. <u>If the carburetor is broken it is not serviced but it costs $500 to replace it</u>. The probability for a carburetor to be broken each time is 5% and is independent across storms. The city has just purchased a new snow plow with its first carburetor.

A. What is the mean total cost to service the first carburetor across all of the storms it is used before it is replaced? Give the <u>best 90% two sided confidence interval</u> for these <u>costs to service the first carburetor</u> before it is replaced.

B. Give the expected mean and standard deviation for total costs to <u>service and replace all carburetors</u> used on this snow plow through the first 30 snow storms